# Replication Plan

Markov State Models from Short Non-Equilibrium Simulations—Analysis and Correction of Estimation Bias

OSTI ID: 1565592

Auto-generated Replication Blueprint

April 8, 2026

## Contents

## Paper Overview

This paper analyzes the bias that arises when Markov state models (MSMs) are estimated from short, non-equilibrium molecular dynamics trajectories. The authors derive analytical expressions for the estimation bias in transition probabilities and relaxation timescales, propose correction methods (e.g., using the Koopman reweighting approach or augmented estimators), and validate their corrections on synthetic multi-state systems and on alanine dipeptide MD data using PyEMMA.

## Detailed Workflow

### Phase 1: Synthetic Test Systems

**1.1 Define synthetic Markov chains.**

- 2-state, 3-state, and multi-well systems with known transition matrices.
- Set timescale separations and equilibrium distributions.

**1.2 Generate non-equilibrium trajectory ensembles.**

- Start trajectories from non-equilibrium initial distributions.
- Vary trajectory length $T$, lag time $\tau$, number of trajectories $M$.

### Phase 2: MSM Estimation

**2.1 Standard MSM estimation.**

- Count transitions $C_{ij}(\tau)$ from trajectory data.
- Estimate transition matrix $T_{ij}(\tau) = C_{ij}(\tau)/\sum_j C_{ij}(\tau)$.
- Compute implied timescales $t_k = -\tau/\ln \lambda_k$.

**2.2 Bias analysis.**

- Compare estimated quantities to exact (known) values for synthetic systems.
- Measure bias as a function of trajectory length and number of trajectories.

### Phase 3: Bias Correction Methods

**3.1 Implement Koopman reweighting.**

- Estimate stationary distribution weights from non-equilibrium data.
- Use reweighted count matrix for MSM estimation.

**3.2 Implement augmented estimator** corrections as described.

**3.3 Compare corrected vs. uncorrected estimates** across parameter regimes.

### Phase 4: Alanine Dipeptide Application

**4.1 Obtain or generate MD trajectory data** for alanine dipeptide.

**4.2 Featurize trajectories** (backbone dihedral angles $\phi$, $\psi$).

**4.3 Cluster into microstates** (e.g., $k$-means on dihedral space).

**4.4 Estimate MSMs** with and without bias correction.

**4.5 Compare implied timescales and free energy surfaces.**

### Phase 5: Validation and Figure Reproduction

**5.1** Reproduce bias-vs-trajectory-length curves.

**5.2** Reproduce implied timescale plots.

**5.3** Reproduce corrected vs. uncorrected comparison figures.

**5.4** Quantitatively validate against paper's reported values.

## Datasets

| Dataset / Source | Type | Location / Notes |
|---|---|---|
| Synthetic Markov chain trajectories | Synthetic | Generated from specified transition matrices |
| Alanine dipeptide MD trajectories | Public / generated | PyEMMA includes a built-in alanine dipeptide dataset; or generate with OpenMM |
| PyEMMA example datasets | Bundled | `http://www.emma-project.org/latest/` (built-in datasets) |

## Models, Tools, and Software

### Programming Languages

- **Python 3.7+** — primary implementation language.

### Libraries and Frameworks

| Tool / Library | Version | Purpose / URL |
|---|---|---|
| PyEMMA | $\geq 2.5$ | MSM estimation, analysis, and visualization `http://www.emma-project.org/` |
| deeptime | latest | Next-gen MSM/Koopman analysis library `https://deeptime-ml.github.io/` |
| NumPy | $\geq 1.18$ | Array operations, linear algebra |
| SciPy | $\geq 1.5$ | Eigenvalue decomposition, statistics |
| Matplotlib | $\geq 3.3$ | Visualization |
| MDTraj | $\geq 1.9$ | MD trajectory I/O and featurization `https://www.mdtraj.org/` |
| OpenMM | $\geq 7.5$ | Optional: MD simulation for alanine dipeptide `https://openmm.org/` |
| scikit-learn | $\geq 0.24$ | Clustering (k-means) |

## Hardware Requirements

- Synthetic systems: any laptop.
- MD simulations: GPU recommended for alanine dipeptide (if generating trajectories).